

INDEPENDENCE, IA, USA

WMO: 720293

Lat: 42.453N

Lon: 91.948W

Elev: 298

StdP: 97.79

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04989

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.6	-20.0	-26.1	0.4	-22.7	-23.0	0.5	-19.6	14.4	-4.1	12.7	-6.0	3.3	310	0.571

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	32.3	23.8	30.9	23.1	29.7	22.0	25.2	30.2	24.1	29.0	23.1	27.5	4.6	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	19.2	28.4	22.5	17.9	26.9	22.0	17.3	26.2	79.1	29.8	73.9	29.1	70.0	27.4	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.0	10.8	9.2	-28.2	35.6	3.2	3.9	-30.6	38.4	-32.5	40.7	-34.3	42.9	-36.6	45.7		
(5)			-28.4	27.3	3.2	1.0	-30.7	28.1	-32.6	28.6	-34.4	29.2	-36.7	29.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.7	-7.1	-5.5	2.2	8.8	15.6	20.8	22.2	20.7	17.4	9.9	2.7	-3.7
	DBStd	11.59	6.95	6.71	6.36	5.38	4.92	3.21	3.10	3.02	4.42	5.16	5.65	6.39
	HDD10.0	2033	529	434	254	85	10	0	0	0	3	66	227	426
	HDD18.3	3874	787	668	501	289	113	12	4	11	69	266	470	684
	CDD10.0	1571	0	0	11	48	183	323	379	332	226	63	7	0
	CDD18.3	370	0	0	1	1	28	85	125	84	42	4	0	0
	CDH23.3	3390	0	0	8	42	303	760	1132	706	389	49	0	0
	CDH26.7	1030	0	0	0	9	90	237	382	191	112	9	0	0
Wind	WSAvg	4.2	4.8	4.8	5.0	5.5	4.9	3.9	2.8	2.5	3.2	3.9	4.7	4.6
Precipitation	PrecAvg	973	29	38	53	95	129	143	122	113	74	66	59	37
	PrecMax	1296	61	115	135	214	271	230	324	256	127	163	148	77
	PrecMin	743	10	6	4	27	24	24	21	21	8	15	9	7
	PrecStd	156	12	25	36	49	54	54	72	65	30	43	40	20
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.0	14.6	24.8	27.6	32.6	33.4	34.1	32.4	32.4	27.6	20.8	12.6
		MCWB	5.5	10.0	16.1	16.5	21.5	23.3	25.4	23.5	22.7	17.5	13.6	9.6
	2%	DB	5.1	7.7	19.0	23.7	29.3	31.3	32.3	30.9	30.1	24.8	16.2	8.7
		MCWB	3.2	5.7	13.2	15.1	19.5	22.5	25.2	23.5	21.5	16.5	10.7	6.3
	5%	DB	2.7	5.0	15.1	21.3	26.4	29.9	30.4	28.9	27.5	22.0	12.9	6.4
		MCWB	1.5	3.0	11.0	13.3	17.5	21.1	23.8	22.6	19.8	15.9	9.1	4.4
	10%	DB	1.2	2.3	11.2	18.0	24.8	27.4	28.8	27.3	25.4	18.9	11.1	3.6
		MCWB	0.6	1.2	8.1	12.0	17.2	19.9	22.7	21.2	18.4	14.4	8.1	2.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.8	10.8	16.8	17.8	22.4	25.6	27.1	25.8	24.6	21.0	14.9	10.2
		MCDB	8.5	14.5	23.2	25.0	29.7	30.9	31.8	30.4	29.7	23.7	18.0	11.3
	2%	WB	3.3	5.8	14.1	16.1	20.7	24.0	25.8	24.6	22.9	18.9	12.2	7.5
		MCDB	5.0	7.8	17.8	22.2	26.9	29.1	30.5	29.1	27.6	22.0	15.5	8.2
	5%	WB	1.7	3.1	11.0	14.5	19.4	22.6	24.6	23.4	21.5	16.5	10.1	4.8
		MCDB	2.6	4.9	14.5	19.9	24.9	27.3	29.5	27.4	25.3	19.7	12.9	6.6
	10%	WB	0.5	1.2	8.2	12.7	18.0	21.5	23.4	22.3	20.3	14.4	7.9	2.6
		MCDB	1.0	2.0	11.2	17.2	22.2	25.6	28.0	26.0	23.9	18.6	10.2	3.6
Mean Daily Temperature Range	5% DB	MDBR	8.8	9.3	9.9	11.8	12.0	11.4	11.6	11.8	13.1	11.8	10.3	8.1
		MCDBR	9.6	11.7	15.5	17.1	14.6	13.7	12.7	12.8	14.6	15.9	14.4	10.4
		MCWBR	7.8	9.0	9.6	9.6	6.7	6.4	6.5	6.8	6.9	8.4	9.5	8.2
	5% WB	MCDBR	9.0	11.6	14.3	14.3	12.8	11.6	11.8	11.1	11.7	12.1	13.6	9.7
		MCWBR	7.4	9.0	9.3	8.9	6.7	6.6	6.7	6.4	6.2	8.4	9.7	8.0
Clear-Sky Solar Irradiance	taub	0.281	0.304	0.330	0.373	0.403	0.419	0.432	0.412	0.382	0.336	0.297	0.279	
	taud	2.385	2.326	2.379	2.288	2.268	2.269	2.217	2.281	2.331	2.456	2.489	2.421	
	Ebn at Noon	866	897	915	894	872	856	840	846	847	852	843	837	
	Edh at Noon	79	99	107	126	133	133	139	126	111	87	71	70	
All-Sky Solar Radiation	RadAvg	1.83	2.85	3.82	4.66	5.55	6.12	6.19	5.39	4.47	2.91	1.96	1.45	
	RadStd	0.18	0.26	0.40	0.40	0.38	0.53	0.38	0.35	0.34	0.36	0.15	0.12	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (53 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page